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Commentary: Cryoanalgesia, a tale of ice and fire?

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CENTRAL MESSAGE

Randomized trials have not shown reduced pain or opioid use with cryoanalgesia in adults undergoing minimally invasive thoracic surgery. This technology should be cautiously used in adult patients.

Cryoanalgesia has been used since the 1970s as a method to reduce postoperative pain.[1] It has been most extensively studied in adult patients undergoing thoracotomy and pediatric patients undergoing pectus excavatum repair. Randomized trials have been promising in these patient populations, with results showing a significant reduction in pain, opioid use, and hospital length of stay associated with cryoanalgesia.[2-5] However, data in adult patients undergoing minimally invasive thoracic surgery by video-assisted thoracoscopic surgery (VATS) or robotic-assisted thoracoscopic surgery is limited. A retrospective study by O'Connor and colleagues[6] suggested cryoanalgesia reduces opioid use in patients undergoing minimally invasive thoracic surgery. Results from the For pain management in post thoRacic procedures via intercOSTal cryoanalgesia (FROST) trial showed no difference in opioid use or hospital length of stay in patients undergoing minimally invasive cardiac surgery.[7] In fact, patients treated with cryoanalgesia had significantly higher pain up through 5 days after surgery compared with the standard of care intercostal nerve block.[7] Our group recently published a randomized trial in patients undergoing pulmonary resection by VATS or robotic-assisted thoracoscopic surgery and found no significant reduction in pain or opioid use in the immediate postoperative period up through the first follow-up visit.[8] Furthermore, the same study found an increased incidence of neuropathic pain 2 weeks after surgery.[8] The literature on cryoanalgesia and postoperative pain is challenging to interpret due to variations in cryotherapy protocols across institutions, treatment techniques, and analgesic methods for comparison. Additionally, primary outcomes, operative procedures, and questionnaires for assessing symptoms, such as pain, vary.

Koliakos and colleagues[9] conducted a randomized trial comparing cryoanalgesia with intercostal nerve block to intercostal nerve block alone in patients undergoing VATS for malignant pulmonary lesions. Following randomization of 40 patients to each treatment group, the authors reported no difference in pain at rest and during coughing in the immediate postoperative period up through 7 days postoperatively. There was no difference in neuropathic pain, Quality of Recovery-15 scores, and opioid consumption during this period. However, 1 month after surgery, pain with coughing was significantly higher in patients treated with cryoanalgesia. There were no differences in pain at 3 and 6 months postoperatively. The authors acknowledge 2 crucial points when interpreting the results. First, the Erbe probe was used off-label for this trial because it is primarily intended for interventional bronchoscopy procedures. Second, complete hypoesthesia of all intercostal spaces was achieved in only 2 patients. This raises the possibility that the cryoanalgesia was not adequate to cause axonal Wallerian degeneration, explaining the lack of effect. Despite these limitations, this study adds to several other studies that suggest a limited influence of cryoanalgesia in adult patients undergoing minimally invasive surgery. Moreover, at the recent American Association for Thoracic Surgery meeting, data were presented suggesting patients may develop long-term neuropathic pain after cryoanalgesia.

Although a few randomized studies showed that cryoanalgesia may reduce pain and opioid use in adult patients undergoing thoracotomy and pediatric patients undergoing pectus excavatum repair, the effect in adult patients is less clear. In

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adult patients, cryoanalgesia appears to be a tale of ice and fire because the postoperative effects may diminish any early benefit.

Conflict of Interest Statement

Dr Weksler serves as a proctor for Intuitive Surgery and is a speaker for AstraZeneca and Merck. The other author reported no conflicts of interest.

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